

Problem: Find all real solutions of $\sqrt{x^2 + y^2} = x + y$

Reasoning Path:

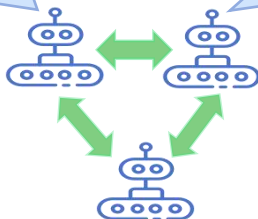
Complete the square on both sides

Identical Reasoning Path

Same mistake

Reasoning Path:

Complete the square on both sides



Reasoning Path:

Complete the square on both sides

Final Answer: $x = 0$ or $y = 0$



Reasoning Path:

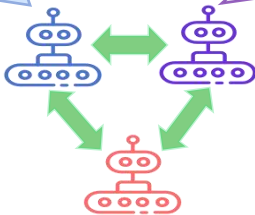
Complete the square on both sides

Distinct Reasoning Path

Maximize solution space exploration

Reasoning Path:

Substitute using polar coordinates



Reasoning Path:

Apply Triangle Inequality Equality

Final Answer: $\{(x, y) \mid x = 0, y \geq 0\} \cup \{(x, y) \mid y = 0, x \geq 0\}$

